# 5.7.3 Compounds and Reactions

**After completing this chapter you should be able to relate properties of elements, compounds and mixtures to scientific models, theories and laws.**

**For Questions 1 to 26 choose the best answer (A), (B), (C) or (D).**

**1.** What is the name given to a chemical that changes colour depending on the acidity of the solution it is in?

(A) Acid.
(B) Base.
(C) Indicator.
(D) Salt.

**2.** What are the products of the reaction between acids and alkalis?

(A) A salt, water and carbon dioxide.
(B) A salt and water.
(C) A salt and hydrogen.
(D) Water and hydrogen.

**3.** A student crushed yellow, red and pink rose petals, added a few drops of water to dissolve any soluble chemicals then filtered the mixture obtained. He then added two drops of each mixture to acidic, basic and neutral solutions. His results are shown in the table.

| Petal colour | Colour in acidic solution | Colour in basic solution | Colour in neutral solution (water) |
|---|---|---|---|
| Pink | Pink | Yellow | Orange |
| Red | Red | Red | Red |
| Yellow | Red | Yellow | Yellow |

Which statement identifies the petals that would make the most useful indicator and the correct reason why?

(A) The pink petals because they change colour in all three types of solutions.
(B) The red petals because they maintain a consistent colour in all solutions.
(C) The yellow petals because their colour in acid and base are very different.
(D) Pink and yellow equally because of the colour changes in acid and base solutions.

**4.** Which of the following is a neutralisation reaction?

(A) Magnesium carbonate + hydrochloric acid.
(B) Diluting concentrated sulfuric acid with water.
(C) Magnesium + hydrochloric acid.
(D) Magnesium hydroxide + sulfuric acid.

**5.** What is the chemical formula for sulfuric acid?

(A) $H_2SO_4$
(B) $H_4SO_2$
(C) $HSO_3$
(D) $HSO_4$

**Use the following information to answer the next TWO questions.**

Sodium hydroxide + nitric acid ⟶ sodium nitrate + X

**6.** What does X represent in this equation?

(A) Carbon dioxide.
(B) Hydrogen.
(C) Sodium.
(D) Water.

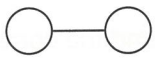
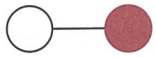
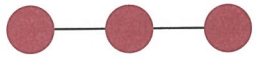
**7.** What type of reaction is this?

(A) Combustion.
(B) Decomposition.
(C) Neutralisation.
(D) Precipitation.

**8.** Which reaction is a combustion reaction?

(A) Adding sulfuric acid to sodium carbonate.
(B) Adding hydrochloric acid to sodium hydroxide.
(C) Burning magnesium metal.
(D) Mixing solutions of silver nitrate and sodium carbonate.

**9.** Which of the following diagrams represents a chemical compound?

(A) 
(B) 
(C) 
(D) 

**10.** Which chemical will react with hydrochloric acid in a neutralisation reaction?

(A) Sodium.
(B) Sodium carbonate.
(C) Sodium chloride.
(D) Sodium hydroxide.

**Use the following information to answer the next FIVE questions.**

| Acidity | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 |
|---|---|---|---|---|---|---|---|---|---|
| Indicator W | Y | Y | Y | O | O | O | R | R | R |
| Indicator X | Y | Y | Y | Y | G | G | G | G | G |
| Indicator Y | C | C | C | C | R | R | R | R | Y |
| Indicator Z | R | R | O | O | O | Y | Y | Y | Y |

**11.** Chemical J had an acidity level of 6. When it was tested with an indicator, the indicator turned green. Which indicator was used?

(A) W
(B) X
(C) Y
(D) Z

**12.** Which indicator(s) would be best to show that a liquid had an acidity level of 5?

(A) Only X.
(B) Only Z.
(C) Both Y and Z used separately.
(D) Either X, Y or Z.

**13.** Three solutions are known to be acidity levels 3, 5 and 9. Which indicators could be added to each to identify the three solutions?

(A) Either W or Y.
(B) Either X or Y.
(C) Either Y or Z.
(D) Either Z or W.

**14.** What colour would indicator X be in water?

(A) Colourless.
(B) Green.
(C) Red.
(D) Yellow.

**15.** Which indicator would seem to be least useful?

(A) W
(B) X
(C) Y
(D) Z

**16.** Consider this chemical reaction:

| | | | | | | |
|---|---|---|---|---|---|---|
| Copper sulfate solution | + | barium nitrate solution | ⟶ | copper nitrate solution | + | barium sulfate solid |

Which choice correctly classifies this reaction?

(A) A combustion reaction.
(B) A decomposition reaction.
(C) A neutralisation reaction.
(D) A precipitation reaction.

**17.** Oxygen and nitrogen molecules in the air can react with each other during lightning storms to produce nitrogen monoxide (NO). Which diagram best represents this reaction?

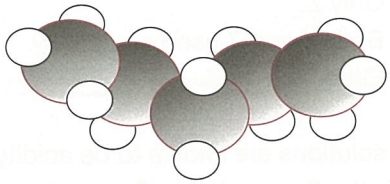
(A) 

(B) 

(C) 

(D) 

**18.** Which choice could show the correct chemical formula for the chemical represented by the diagram?



(A) $C_5H_{10}$
(B) $C_5O_7H_5$
(C) $5CO_2$
(D) $C_5H_{12}$

**19.** A molecule of sodium carbonate contains atoms of sodium, carbon and oxygen in the ratio 2:1:3. Which chemical formula for sugar is correct?

(A) $S_2C_1O_3$
(B) $Na_2C_20_3$
(C) $Na_2CO_3$
(D) $NaC_2O_3$

**20.** What is the main role of indicators?

(A) To identify acids and bases.
(B) To determine when neutralisation reactions end.
(C) To neutralise acids and bases.
(D) To make acids and bases.

**21.** Which of the following is a decomposition reaction?

(A) Copper carbonate solid + heat → copper oxide solid + carbon dioxide
(B) Hydrochloric acid + copper carbonate → copper chloride + carbon dioxide
(C) Copper oxide solid + water → copper hydroxide
(D) Hydrogen gas + oxygen gas→ water

**Use the following information to answer the next THREE questions.**

| Compounds | Reaction with dilute sulfuric acid | Reaction when heated in air |
|---|---|---|
| E | Forms a salt, water and carbon dioxide. | Forms an oxide and carbon dioxide. |
| F | No reaction. | No reaction. |
| G | Forms a salt and hydrogen gas. | Forms an oxide and water. |
| H | Forms a salt and water. | Forms an oxide and water. |
| I | Forms a salt, water and carbon dioxide. | Forms an oxide and carbon dioxide. |
| J | Forms a salt and water. | No reaction. |
| K | No reaction. | No reaction. |
| L | Forms a salt and hydrogen gas. | No reaction. |
| M | Forms a salt, water and carbon dioxide. | Forms an oxide and carbon dioxide. |

**22.** Which compounds are the most similar?

(A) E and G.
(B) E and I.
(C) F and K.
(D) H and J.

**23.** Which compound could be copper hydroxide?

(A) G
(B) H
(C) I
(D) J

**24.** Which equation best represents the reaction between I and sulfuric acid?

(A) Copper carbonate + sulfuric acid → copper sulfate + water + carbon dioxide
(B) Copper hydroxide + sulfuric acid → copper sulfate + water + carbon dioxide
(C) Copper chloride + sulfuric acid → copper sulfate + water + carbon dioxide
(D) Copper sulfate + sulfuric acid → copper sulfate + water + carbon dioxide

**25.** When lead oxide is heated strongly with carbon, the lead oxide will lose oxygen and metallic lead will be formed. The oxygen will combine with the carbon to form carbon monoxide. Which diagram represents this reaction?

**Key**

● = Carbon ○ = Oxygen ■ = Lead

(A) 

(B) 

(C) 

(D) 

**26.** Which of the following compounds is not an acid?

(A) HCl
(B) $H_2SO_4$
(C) $H_2SO_3$
(D) $H_2O_2$

**27.**

(a) What is a precipitate?

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(b) How can we tell if a precipitate forms in a test tube?

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(c) How can we separate a precipitate from the rest of the contents in the test tube in which it forms?

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(d) Why can we separate it this way?

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**28.** The table shows the results of an experiment set up to examine factors affecting the rusting of iron.

| Tube | Contents of tube | Observations after five days |
|---|---|---|
| A | Iron nail. | Nail unaffected. |
| B | Iron nail, water, nail half-exposed. | Nail in water coloured by thin rust layer, nail at boundary of air and water covered with thick rusty flakes. |
| C | Iron nail, boiled water, nail half-exposed. | Nail in water slightly coloured by rust, nail at boundary of air and water covered with thick rusty flakes. |
| D | Iron nail, boiled water, nail fully immersed, layer of oil on top of water to prevent air redissolving in the water. | Nail unaffected. |
| E | Iron nail, plug of cotton wool above nail, crystals of calcium chloride on plug to absorb moisture from the air. | Nail unaffected. |

(a) State four factors that would need to be controlled if this experiment was a 'fair test'.

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(b) Which tube could be regarded as the 'control' tube? Explain why.

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(c) What was the purpose of boiling the water in test tubes C and D?

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(d) What factors are affecting the rusting of irons?

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(e) Write a conclusion for the experiment.

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**29.** What is the name given to the reaction between an acid and a base?

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**30.**

(a) What is a chemical indicator?

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(b) State two uses for chemical indicators in the laboratory.

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(c) Outline one commercial use of chemical indicators.

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**31.** Write word equations for the chemical reaction between each of the following:

(a) Hydrochloric acid and sodium hydroxide

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(b) Hydrochloric acid and magnesium hydroxide

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(c) Sulfuric acid and lead hydroxide

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(d) Nitric acid and tin hydroxide

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(e) Carbonic acid and potassium hydroxide

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**32.**

(a) What type of chemical would you expect to find in an *antacid* tablet?

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(b) Explain why.

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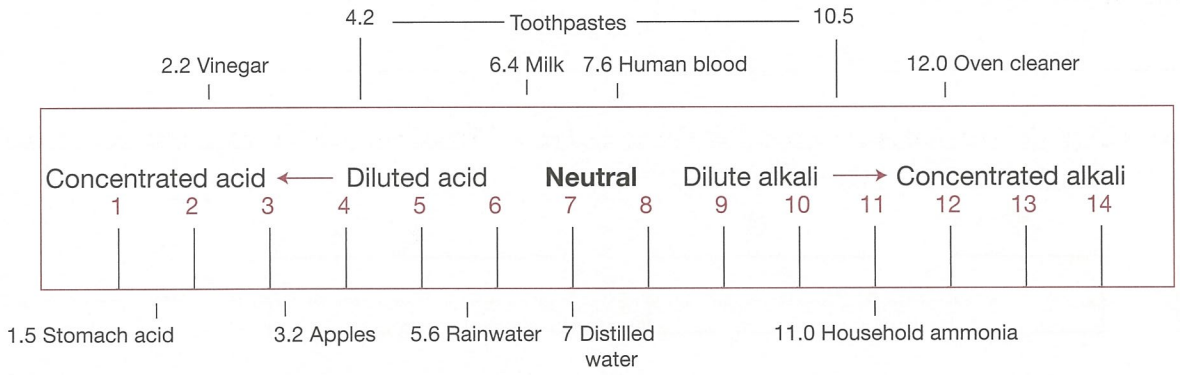
(c) Explain how neutralisation is the process that eases indigestion.

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**33.** Use the pH scale diagram below to answer the questions.




(a) Which substance shown is most acidic?

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(b) Which substance is most alkaline?

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(c) Which substance is least acidic?

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(d) Which is least alkaline?

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(e) What does a pH of 7 indicate about a substance?

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(f) Why does rainwater not have a pH of 7?

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(g) Which is the stronger alkali, household ammonia or oven cleaner? Justify your answer.

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(h) In which region of the pH scale would you find indigestion tablets?

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(i) Explain your answer to (g).

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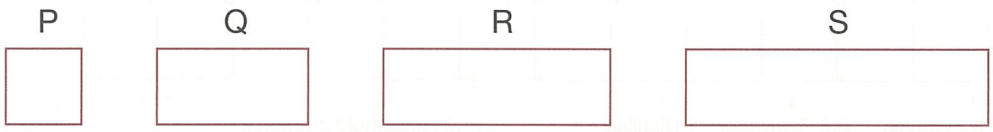
(j) Most 'soft drinks' are acidic. What acid do they contain?

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(k) How does this acid get there?

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**34.** A student was given four pieces of magnesium ribbon as shown. These were to be used in an experiment in which the magnesium was reacted with hydrochloric acid.




(a) Write a word equation for the reaction between magnesium and the acid.

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(b) After doing the experiment, the students concluded 'The time it takes for magnesium to react with acid is proportional to the mass of magnesium used.' Explain what this conclusion means.

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(c) Predict the observations made by the students which led them to this conclusion.

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(d) The conclusion the students wrote (see b) is incorrect. Rewrite it so that it reflects the design of the experiment more accurately.

(e) Explain why it is unnecessary to find the mass of each piece of magnesium for this experiment.

(f) Name *two* variables that would need to be controlled if this experiment was a fair test.

(g) Is a control (a test tube with acid but no magnesium) needed in this experiment? Explain your answer.

(h) How could the experiment be improved?

**35.** Some hydrochloric acid was added to four different chemical compounds in different test tubes. The results are shown in the table.

| | Volume of gas given off (mL) | | | |
|---|---|---|---|---|
| **Time (mins)** | **Calcium carbonate** | **Copper carbonate** | **Magnesium carbonate** | **Zinc carbonate** |
| 1 | 1 | 2 | 1 | 1 |
| 2 | 3 | 5 | 4 | 2 |
| 3 | 7 | 9 | 8 | 5 |
| 4 | 11 | 16 | 13 | 9 |

(a) According to this data, which compound was the fastest to react with the acid?

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(b) Justify your answer to (a).

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(c) Which of the following four factors *does not* have to be controlled for this to be a fair test of which compound reacted fastest?

(A) The concentration of the acid.
(B) Whether the test tubes are shaken or not.
(C) The mass of each compound used.
(D) The size of the test tube rack.

(d) Which statement about the relative chemical activity of these compounds is true?

(A) The activity of calcium carbonate is greater than the activity of copper carbonate.
(B) The activity of zinc carbonate is greater than the activity of calcium carbonate.
(C) The activity of magnesium carbonate is greater than the activity of zinc carbonate.
(D) The activity of zinc carbonate is greater than the activity of copper carbonate.

(e) Name the gas given off during these reactions.

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(f) Write the chemical formula for this gas.

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(g) Write the chemical formula for hydrochloric acid.

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**36.** Some students tied a balloon over the end of a flask that contained hydrochloric acid and sodium carbonate. The results of the experiment they did are summarised in the table. Use them to answer this question.

| Flask | Volume of acid (mL) | Mass of sodium carbonate (g) | Temperature of contents (°C) | Time for all carbonate to be used (s) | Circumference of balloon at end of time (cm) |
|---|---|---|---|---|---|
| 1 | 25 | 2.0 | 30 | 15 | 18 |
| 2 | 25 | 2.0 | 10 | 45 | 18 |
| 3 | 25 | 2.0 | 20 | 30 | 18 |

(a) What was being *investigated* in this experiment?

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(b) If this experiment was a fair test, what factors were controlled?

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(c) What was being *measured* in this experiment?

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(d) Why was the circumference of the balloon measured?

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(e) Explain how this measurement contributes to the purpose of the experiment.

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(f) What do the results of the experiment show?

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(g) Why was the circumference of the balloon the same at the end of each time?

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(h) Write a conclusion for the experiment.

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**Notes**